Summary in Graph

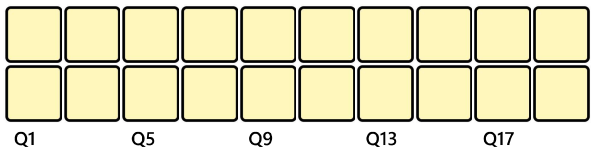
Exam Summary (GO Classes CS&DA Test Series 2026 | Linear Algebra | Topic Wise Test 1)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**20** (10 + 10)
Total Marks:	**30** (10 + 20)
Exam Duration:	**60 Minutes**
Time Taken:	**60 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (199) out of 1986 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let v_1, v_2, w be vectors in $\mathbf{R}^3$, and suppose that the matrix whose three columns are v_1, v_2, and w has three pivots. Which of the following statements must be true? Clearly circle all that apply.

A. Every vector in $\mathbf{R}^3$ is a linear combination of v_1, v_2, and w.
B. w is not a linear combination of v_1 and v_2.
C. Every vector in $\mathbf{R}^2$ is a linear combination of v_1 and w.

D. Every vector in $\mathbf{R}^2$ is a linear combination of v_1 and v_2.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $A = \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix}$. Find A^{-1}.

A. $A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix}$

B. $A^{-1} = \frac{1}{-2}\begin{pmatrix} 1 & -1 \\ -2 & 4 \end{pmatrix}$

C. $A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -2 & 4 \end{pmatrix}$

D. $A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -2 \\ -1 & 4 \end{pmatrix}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are linearly independent sets ?

A. $\left\{ \begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix} \right\}$

B. $\left\{ \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$

C. $\left\{ \begin{bmatrix} 4 \\ -6 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 6 \\ -9 \\ 3 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 2 \\ 0 \end{bmatrix} \right\}$

D. $\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} \right\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are TRUE ?

A. If $\det(A) = 0$ then columns of A are linearly dependent.
B. Let $v_1, \ldots, v_m$ be eigenvectors of A with eigenvalues $\lambda_1, \ldots, \lambda_m$. Suppose that $\lambda_1, \ldots, \lambda_m$ are distinct then $v_1, \ldots, v_m$ are linearly independent.
C. If a given linear system has 5 unknowns x_i and 7 equations, then the system must be inconsistent.
D. A square matrix A is invertible if and only if 0 is not an eigenvalue of A.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Suppose that A is an $m \times n$ matrix with entries in $\mathbb{R}$ and consider a system of linear equations $Ax = b$. Which of the following is/are TRUE ?

A. If $\text{rank}(A) = m$ and $n > m$ then the system $Ax = b$ has infinitely many solutions.
B. If $\text{rank}(A) = n$, then $Ax = b$ has a unique solution.
C. If $\text{rank}(A) = m$ and $n \geq m$, then $Ax = b$ has at least one solution.
D. If $\text{rank}(A) = m$ and $n = m$, then $Ax = b$ could have no solution.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following statements are true for all $n \times n$ matrices A, B :

A. $\left(A^T\right)^T = A$;
B. $\left|A^T\right| = |A|$;
C. $(AB)^T = A^T B^T$;
D. $(A + B)^T = A^T + B^T$.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following statements are true?

A. Let A be a $n \times m$ matrix with $n < m$. Then there is a nonzero solution y with $Ay = 0$ only if A has full row rank.
B. Let A be a $n \times m$ matrix with $n > m$. There is a nonzero solution y with $Ay = 0$.
C. The rank of a matrix is equal to the maximum number of linearly independent rows in the matrix.
D. Let A be an $n \times n$ matrix. Suppose $A = BC$, where B has size $n \times r$ and C has size $r \times n$. The rank of A is less than or equal to r.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are $\mathbf{TRUE}$?

A. If A and B are 4×4 matrices such that $\text{rank}(AB) = 3$, then $\text{rank}(BA) < 4$.
B. If A is a 5×3 matrix with $\text{rank}(A) = 2$, then for every vector $b \in \mathbb{R}^5$ the equation $Ax = b$ will have at least one solution.
C. If A is a 4×7 matrix, then A and A^T have the same rank.
D. Let A and $B \neq 0$ be 2×2 matrices. If $AB = 0$, then A must be the zero matrix.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a 4×4 matrix with determinant 7. Which of the following is/are TRUE ?

A. For some vector $\mathbf{b}$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution.
B. For some vector $\mathbf{b}$ the equation $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.
C. For some vector $\mathbf{b}$ the equation $A\mathbf{x} = \mathbf{b}$ has no solution.
D. For all vectors $\mathbf{b}$ the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are TRUE ?

A. If A is an invertible 4×4 matrix, then $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$, where A^T denotes the transpose of A.
B. If A and B are 3×3 matrices with $\operatorname{rank}(A) = \operatorname{rank}(B) = 2$, then $\operatorname{rank}(AB) = 2$.
C. If A and B are invertible 3×3 matrices, then $A + B$ is invertible.
D. If A is an $n \times n$ matrix with rank less than n, then for any vector $\mathbf{b}$, the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following is/are TRUE?

A. If the columns of matrix A are linearly dependent, then $\mathrm{Ax} = 0$ has a nontrivial solution.
B. If A is a $m \times r$ matrix with r independent columns and B is a $r \times n$ matrix with r independent rows, then AB is invertible.
C. For a matrix A, let λ_1 and λ_2 be two different eigenvalues with corresponding eigenvectors x_1 and x_2. Then, $x_1 + x_2$ is an eigenvector for A.
D. There exists a 4×3 real matrix A and a 3×4 real matrix B such that the column vectors of AB are linearly independent.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be a 3×5 matrix A and $\mathbf{b}$ in $\mathbb{R}^3$. Which of the following statements about the matrix equation $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x} \in \mathbb{R}^5$, and the corresponding homogeneous equation $A\mathbf{x} = \mathbf{0}$, could be true?

A. $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.
B. $A\mathbf{x} = \mathbf{0}$ is inconsistent.
C. $A\mathbf{x} = \mathbf{0}$ has exactly two solutions.
D. $A\mathbf{x} = \mathbf{0}$ has a unique solution and $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the following linear system.

$$\begin{aligned} x + ay + z &= a + 1 \\ 2x + by + z &= b + 1 \\ 3x + cy + z &= c + 1 \end{aligned}$$

Suppose the system only has unique solution. Then

A. $x = 1$
B. $y = 0$
C. $z = 1$
D. $x = 2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let

$$A = \begin{pmatrix} 0 & 0 & 0 & a \\ -1 & 0 & 0 & b \\ 0 & -1 & 0 & c \\ 0 & 0 & -1 & d \end{pmatrix}$$

Suppose 0 is an eigenvalue of A with multiplicity 2 . Then which of the following statement is correct?

A. $a = 0, b \neq 0$
B. $a = b = 0$ and $c \neq 0$
C. $a = c = 0$ but $b \neq 0$.
D. $a = d = 0$ but $b \neq 0$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 15min 48sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following is/are TRUE ?

A. If a system $Ax = b$ has more than one solution, then so does the system $Ax = 0$.
B. If $BC = BD$, then $C = D$.
C. If A and B are $n \times n$ matrices, then $(A + B)(A - B) = A^2 - B^2$.
D. If A is invertible and $r \neq 0$, then $(rA)^{-1} = rA^{-1}$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

. If the matrix $A = \begin{pmatrix} a & 1 \\ 2 & 3 \end{pmatrix}$ has 1 as an eigenvalue, then the determinant of A is

A. 5

B. 2

C. 4

D. 3

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The rank of the matrix

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 1 & 0 & a & 0 \\ 1 & 0 & 0 & 0 & b \end{pmatrix}.$$

A. depends on the values of both a and b
B. is independent of the values of both a and b
C. depends on the value of a but not on the value of b
D. depends on the value of b but not on the value of a

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

An $n \times n$ matrix $A = (a_{ij})$ is said to be antisymmetric if $a_{ij} = -a_{ji}$ for all $1 \leq i, j \leq n$. Which of the following is/are true?

A. If A is antisymmetric then all its principal diagonal entries are zero.
B. If A, B are antisymmetric matrices then the product AB need not be antisymmetric.
C. If A is antisymmetric then A^2 is symmetric.
D. Every nonzero 3×3 antisymmetric matrix is invertible.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Suppose A is a symmetric $n \times n$ matrix. If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are vectors satisfying $A\mathbf{x} = 2\mathbf{x}$ and $A\mathbf{y} = 3\mathbf{y}$, then what is the value of dot product between $\mathbf{x}$ and $\mathbf{y}$ i.e. $x.y$?

A. 0
B. -1
C. 1
D. 6

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be an $m \times n$ matrix and x and b are column vectors.
Which of the following is/are TRUE?

A. If $Ax = 0$ has only the trivial solution $x = 0$, then $Ax = b$ always has a unique solution.
B. There are vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3 \in \mathbb{R}^m$ such that $A\mathbf{x} = \mathbf{b}_1$ has no solutions, $A\mathbf{x} = \mathbf{b}_2$ has one solution, and $A\mathbf{x} = \mathbf{b}_3$ has infinitely many solutions.
C. If the columns of A are linearly independent, then the equation $Ax = b$ has exactly one solution for every b.
D. If a system $Ax = b$ has more than one solution, then so does the system $Ax = 0$.

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

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